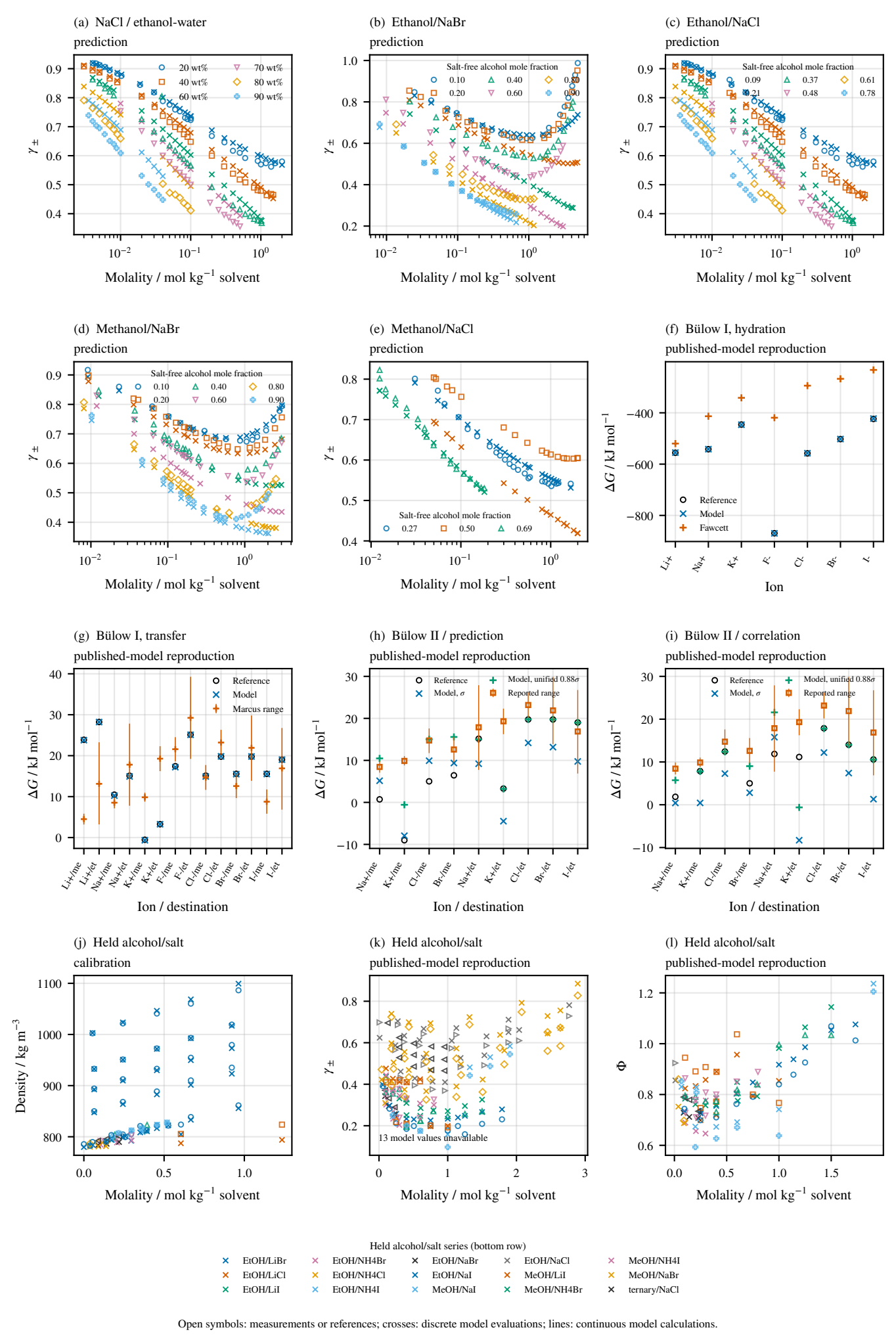

(a) NaCl / ethanol-water
prediction
20 wt%
40 wt%
60 wt%
70 wt%
80 wt%
90 wt%
γ ±
Molality / mol kg^{-1} solvent
(b) Ethanol/NaBr
prediction
Salt-free alcohol mole fraction
0.10
0.20
0.40
0.60
0.80
0.90
γ ±
Molality / mol kg^{-1} solvent
(c) Ethanol/NaCl
prediction
Salt-free alcohol mole fraction
0.09
0.21
0.37
0.48
0.61
0.78
γ ±
Molality / mol kg^{-1} solvent
(d) Methanol/NaBr
prediction
Salt-free alcohol mole fraction
0.10
0.20
0.40
0.60
0.80
0.90
γ ±
Molality / mol kg^{-1} solvent
(e) Methanol/NaCl
prediction
Salt-free alcohol mole fraction
0.27
0.50
0.69
γ ±
Molality / mol kg^{-1} solvent
(f) Bülow I, hydration
published-model reproduction
Reference
Model
Fawcett
ΔG / kJ mol^{-1}
Li+
Na+
K+
F-
Cl-
Br-
I-
Ion
(g) Bülow I, transfer
published-model reproduction
Reference
Model
Marcus range
ΔG / kJ mol^{-1}
Li+/me
Li+/et
Na+/me
Na+/et
K+/me
K+/et
F-/me
F-/et
Cl-/me
Cl-/et
Br-/me
Br-/et
I-/me
I-/et
Ion / destination
(h) Bülow II / prediction
published-model reproduction
Reference
Model, σ
Model, unified 0.88σ
Reported range
ΔG / kJ mol^{-1}
Na+/me
K+/me
Cl-/me
Br-/me
Na+/et
K+/et
Cl-/et
Br-/et
I-/et
Ion / destination
(i) Bülow II / correlation
published-model reproduction
Reference
Model, σ
Model, unified 0.88σ
Reported range
ΔG / kJ mol^{-1}
Na+/me
K+/me
Cl-/me
Br-/me
Na+/et
K+/et
Cl-/et
Br-/et
I-/et
Ion / destination
(j) Held alcohol/salt
calibration
Density / kg m^{-3}
Molality / mol kg^{-1} solvent
(k) Held alcohol/salt
published-model reproduction
13 model values unavailable
γ ±
Molality / mol kg^{-1} solvent
(l) Held alcohol/salt
published-model reproduction
Φ
Molality / mol kg^{-1} solvent
Held alcohol/salt series (bottom row)
EtOH/LiBr
EtOH/LiCl
EtOH/LiI
EtOH/NH4Br
EtOH/NH4Cl
EtOH/NH4I
EtOH/NaBr
EtOH/NaI
MeOH/NaI
EtOH/NaCl
MeOH/LiI
MeOH/NH4Br
MeOH/NH4I
MeOH/NaBr
ternary/NaCl
Open symbols: measurements or references; crosses: discrete model evaluations; lines: continuous model calculations.